Aim: Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

Tasks Performed:

- Opened the ChronoVault DApp and selected MetaMask on the Connect Wallet screen.
- Reviewed the MetaMask connection request and the permissions requested by the website.
- Confirmed that the requested network was Sepolia and approved the connection.
- Verified the connected wallet address and Connected status on the dashboard.
- Entered the ChronoVault account details (name, Ethereum address and 20-digit secret) in the Add Account form.
- Opened the wallet deployment workflow and reviewed the smart-wallet deployment request in MetaMask.

Theory:

Cryptocurrency Wallet – A wallet provides an interface for managing blockchain accounts and approving transactions. It holds the user's keys and lets the user decide which transactions are signed and sent.

MetaMask – A browser wallet that connects a decentralized application (DApp) to an Ethereum-compatible network and exposes the connected account to the application. The DApp can request permission to view the account and to send transaction requests, while MetaMask keeps control of approval, so funds cannot be moved without the user's consent.

DApp Integration – Integration generally involves connecting the wallet, selecting an enabled blockchain network, and approving the permissions requested by the application.

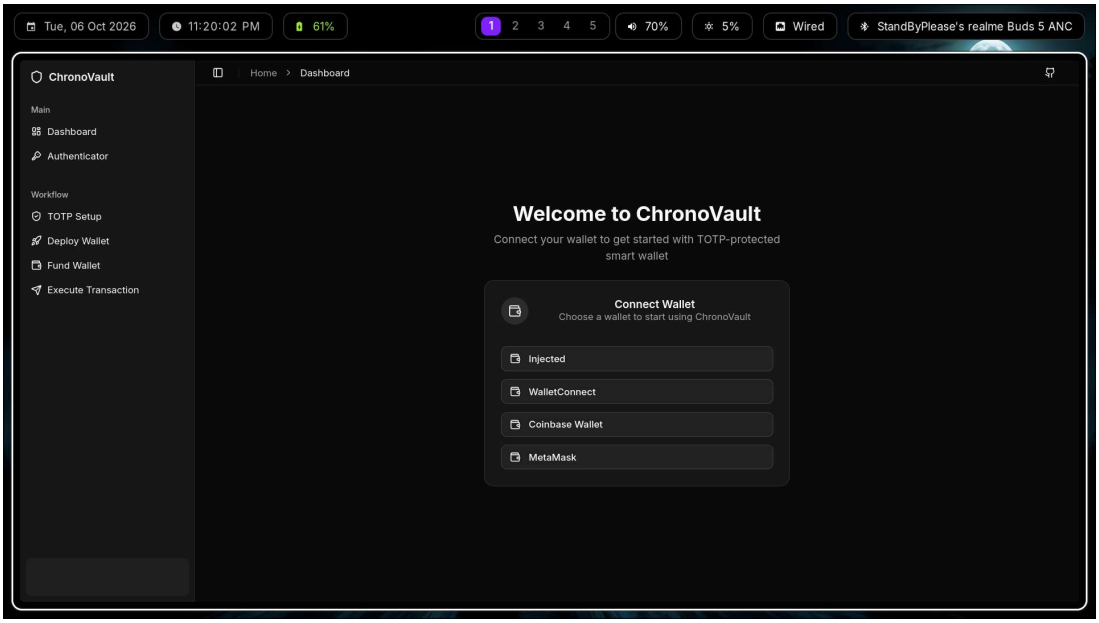
ChronoVault DApp – In this experiment the ChronoVault DApp uses MetaMask to connect an account. It offers workflow options such as TOTP setup, wallet deployment, funding and transaction execution. The screenshots show the connection request, the network permission confirmation, the connected-wallet dashboard, the account details form and a deployment request on the Sepolia network.

Code & Output:

Step 1 — Open the ChronoVault DApp and view the wallet options

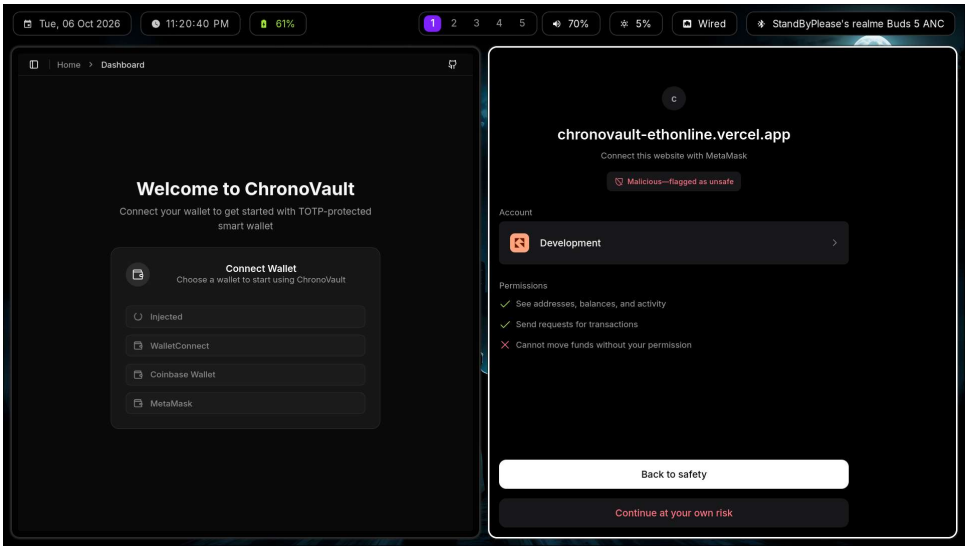
https://chronovault-ethonline.vercel.app

The welcome screen offers Injected, WalletConnect, Coinbase Wallet and MetaMask as connection options.



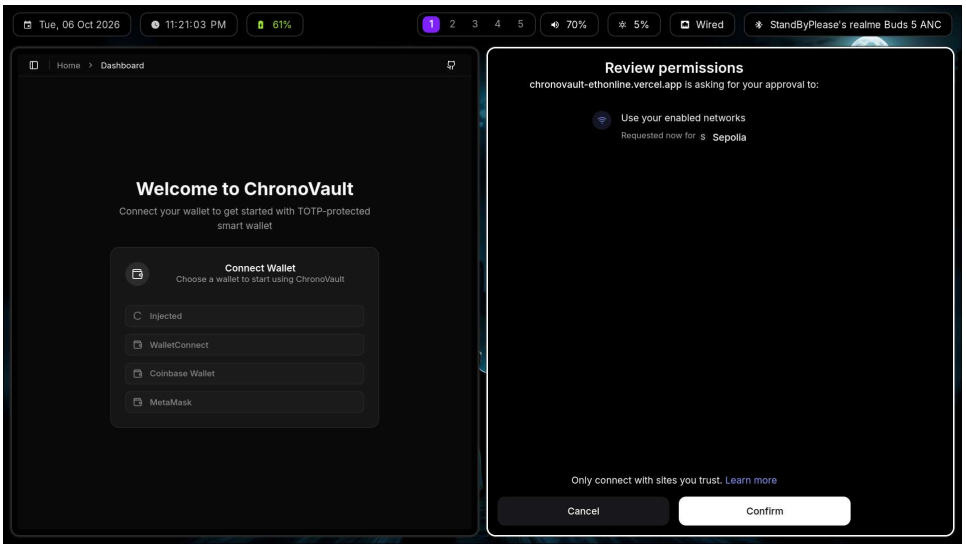
Step 2 — Select MetaMask and review the connection request

MetaMask shows the connection request for chronovault-ethonline.vercel.app with the selected account. The permissions listed are to see addresses, balances and activity and to send requests for transactions, and it cannot move funds without permission.



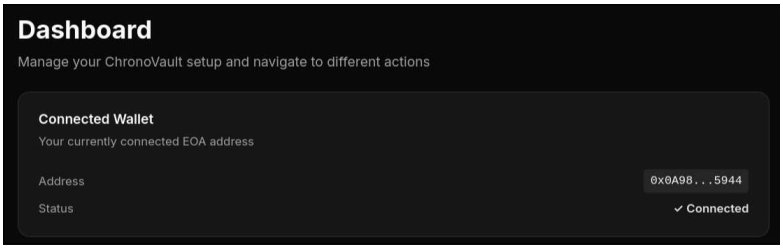
Step 3 — Review the network permission

The site asks to use the enabled networks, and the requested network is Sepolia. The connection is approved with Confirm.



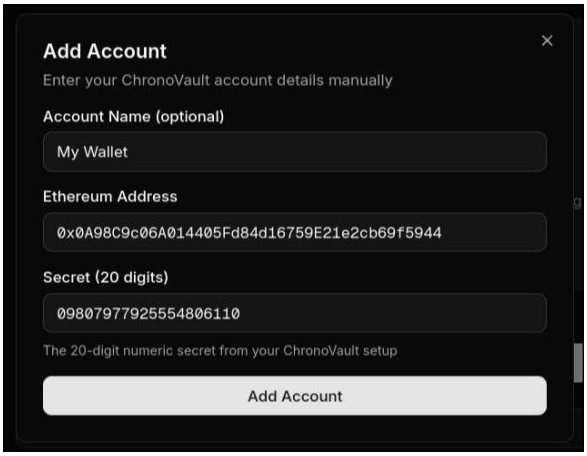
Step 4 — Verify the connected wallet on the dashboard

The dashboard shows the connected EOA address and the status Connected.



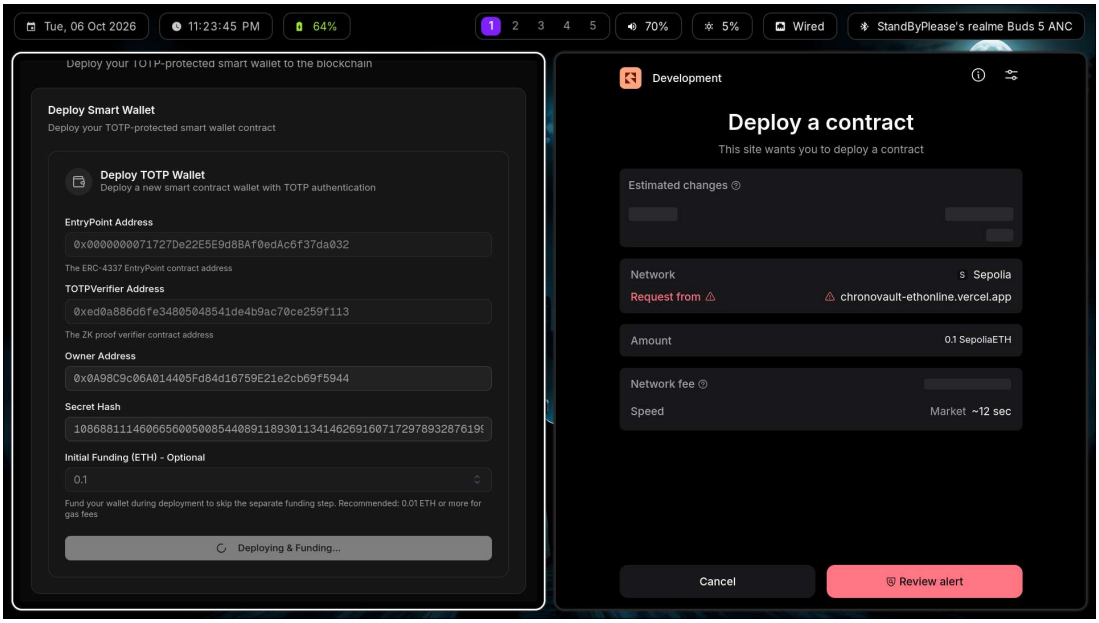
Step 5 — Enter the ChronoVault account details

In the Add Account form, the account name, Ethereum address and the 20-digit secret from the ChronoVault setup are entered.



Step 6 — Review the smart wallet deployment request

The Deploy TOTP Wallet workflow lists the EntryPoint, TOTPVerifier and owner addresses, the secret hash and the 0.1 ETH initial funding. MetaMask then shows a “Deploy a contract” request on Sepolia for review.



Conclusion:

The **MetaMask** wallet was connected to the **ChronoVault DApp** and the requested permissions were reviewed for the **Sepolia** network. The dashboard displayed the connected wallet address and status, the account configuration form was reached successfully, and the deployment workflow presented a smart-wallet deployment request through MetaMask. This demonstrated the **integration of a browser cryptocurrency wallet with a decentralized application**.